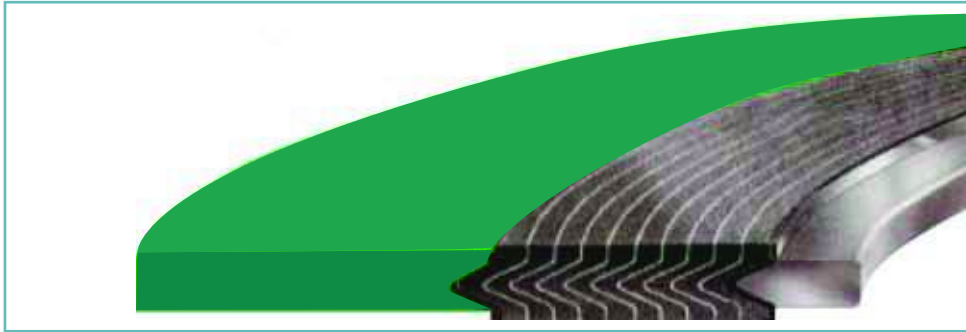




General information and description

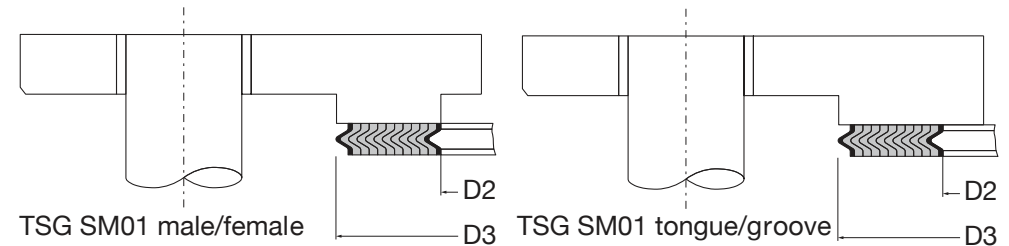
The sealing element of the spiral wound gasket consists of a v-shaped metal strip spirally wound in combination with a soft sealing material filler. The metal strip provides outstanding resilience, while the flexible sealing filler guarantees excellent sealing. Due to this combination of materials the spiral wound gasket is suitable for sealing under severely fluctuating temperature and pressure conditions. Depending on the application the spiral wound gasket can be specified with outer and/or inner rings.

Characteristics

- The spiral wound gasket is suitable for use across a wide pressure range and is therefore virtually applicable.
- The spiral wound gasket can be used to seal fluid pressures up to 250 bar and damage (although extra care should be taken during transportation and installation of large diameter gasket without guide rings).
- The outer guide ring serves to locate the spiral element centrally on the flange faces and prevent blow-out.
- By combining different winding materials and metals the gasket can be tailored to a wide variety of operating conditions.
- Due to its non-adhesive character the gasket is easy to remove after service.
- The gasket does not cause damage to the flange faces

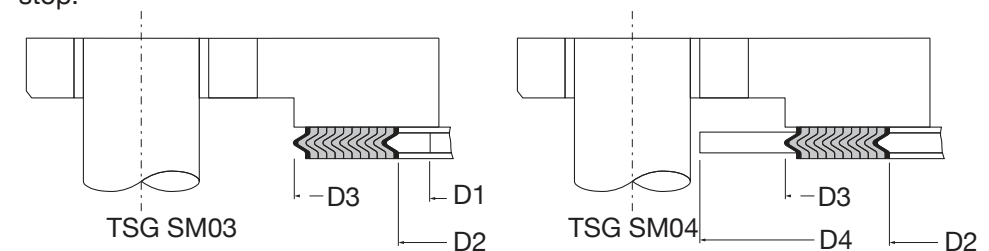
TSG SM01 SWG without guide or inner ring

This style of gasket is mainly used for male and female flanges or projection and recess flanges and used in load-bearing flange configurations, such as tongue and groove flanges. Standard dimensions according to EN/ASME.



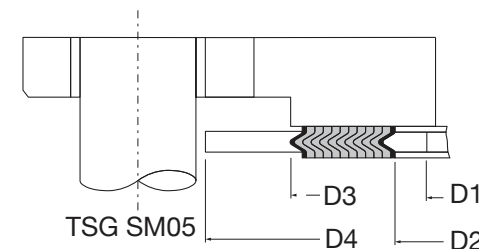
TSG SM03 SWG with inner ring TSG SM04 SWG with outer guide ring

Mainly used for male and female flanges or projection and recess flanges. The inner ring is necessary as a compression stop. Standard type for RF flange connections.



TSG SM05 SWG with inner and outer guide ring

This style offers the best performance for raised-face flange connections.



Benefits of the centring ring

The spiral wound gasket outer centring ring provides the following benefits:

- Optimum location between the bolts.
- Protection of the spiral wound element.
- Additional security against gasket blow-out.
- Acts as compression limiter preventing overloading and overcompression of the spiral wound element.
- Prevents radial-flow of soft fillers, such as P.T.F.E.

For these reasons it is preferable to use spiral wound gasket with outer centring rings. The the outer ring is marked with nominal size, pressure class, standard and materials.

Benefits of the inner ring

The spiral gasket inner ring provides the following benefits:

- Prevents radial-flow of soft fillers, such as P.T.F.E.
- Reduces turbulence-minimising flow resistance and crevice corrosion.
- Acts as a heat shield when the spiral wound gasket is subjected to high temperatures.
- Avoidance of over-compression at elevated seating loads in high-pressure service
- Improvement of the load distribution over the gasket

Inner and outer rings are particularly recommended for use on spiral wound askets exceeding class 600lbs, but specifically recommended for high temperatures and Pressures to optimise the operational reliability of the spiral wound sealing element.

Material selection

Material selected for the inner ring and winding metal is usually the same as the flange metal, in order to prevent corrosion and differential expansion problems. The outer centring ring is generally manufactured from carbon steel with an anti-corrosion treatment, however the ring may also be manufactured in the same metal as the flange to prevent corrosion.

Ring materials

Material	DIN specification	DIN Nr.	AISI/ ASTM UNS	B.S.	Temp. [°C] min max	
Carbon steel	RSt. 37.2 CS	1.0038	238-C	40B	-40	500
SS 304	X5CrNi 18	1.4301	304	304S15/16/31	-250	550
SSI 304L	X2CrNi 189	1.4306	304L	304S11	-250	550
SS 309	X15CrNiSi 2012	1.4828	309	309S24	-100	1000
SSI 316	X5CrNiMo 1810	1.4401	316	316S31/33	-100	550
SS 316L	X2CrNiMo 1810	1.4404	316L	316S11/13	-100	550
SS 316Ti	X10CrNiMoTi 1810	1.4571	316Ti	320S31	-100	550
SS 321	X10CrNiTi 189	1.4541	321	321S12/49/87	-250	550
SS 347	X6CrNiNb 1810	1.4550	347	347S31	-250	550
Monel 400	NiCu 30 Fe	2.4360	NO4400	3072-76NA13	-125	600
Inconel 600	NiCr 15 Fe	2.4816	NO6600	3072-76NA14	-100	950
Incoloy 800	X10NiCrAlTi 3220	1.4876	NO8800	3072-76NA15	-100	850
Incoloy 825	NiCr 21 Mo	2.4858	NO8825	3072-76NA16	-100	450
Hastelloy B2	NiMo 28	2.4617	N10665	---	-200	450
Hastelloy C276	NiMo16Cr15W	2.4819	N10267	---	-200	450
Titanium (gr.1)	Ti 99,8	3.7025	---	---	-250	350

The table below may be used for determining the correct filler. It should be pointed out that graphite is the optimum filler in many cases. Only when graphite would cause medium pollution, or is not chemically resistant, is it advisable to apply another type of filler material.

Standard fillers

Material	Temp. [°C] Min. Max.		Max. operating press. [MPa]	Gastight- ness	Application	Colour code
Graphite	-200	550	25	Good	aggressive medium	Grey
PTFE	-200	250	10	Good	aggressive medium	White

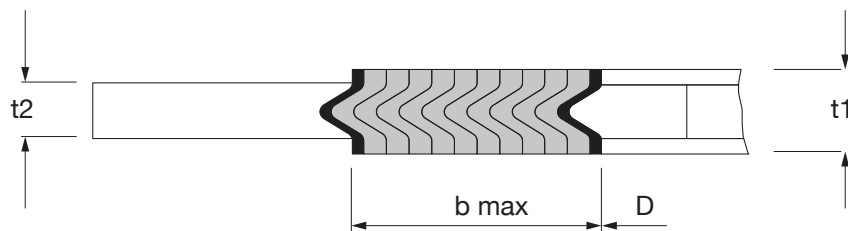
Graphite is universally applicable, high-quality, asbestos-free material with a very good chemical resistance, resistance to ageing, good gastightness.

PTFE is a high quality synthetic material with an excellent chemical resistance, resistance to temperature up to 250°C, resistant to ageing, excellent gastightness.

Non standard materials are available on request.

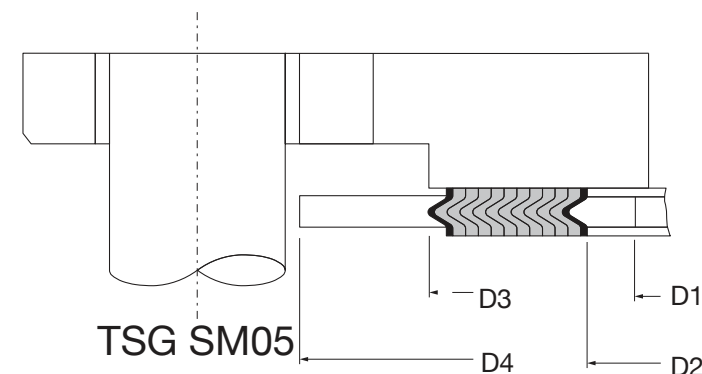
Selection of gasket thickness

Spiral wound gasket is standard produced with a thickness of 4,5mm. However, in some cases another thickness can be selected to obtain effective sealing. Please see the table below:



t1 [mm]	Tolerance [mm]	D [mm]	b max [mm]	t2	Thickness after fitting [mm]
7.2	+ 0,35	100 - 3200	33	5	5.3 - 5.6
6.4	+ 0,30	100 - 1600	30	4	4.7 - 4.9
6.4	+ 0,30	1601 - 3200	25	4	4.7 - 4.9
4.8	+ 0,25	15 - 630	35	3	3.2 - 3.4
4.8	+ 0,25	631 - 1600	30	3	3.2 - 3.4
4.8	+ 0,25	1601 - 2000	20	3	3.2 - 3.4
3.5	+ 0,25	15 - 1000	25	2	2.3 - 2.5
3.2	+ 0,25	15 - 1000	20	3	2.3 - 2.5
2.5	+ 0,25	15 - 500	10	1.5	1.8 - 2.0

Production tolerances (pt)

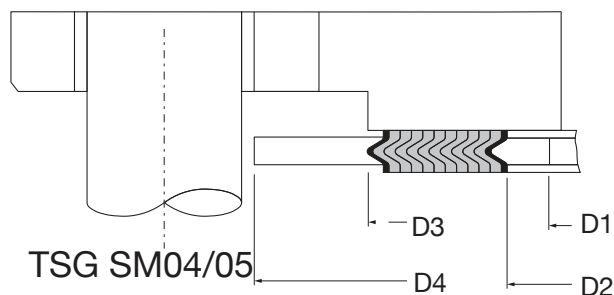


Gasket Standard	ASME B 16.20		BS 3381		DIN	
Flange Standard	ASME/ANSI B16.5	ASME/ANSI B16.47 series A,B	BS 1560	DIN		
	Dim.	Dia.	Dia.	PT [mm]	Dia.	PT[mm]
D1	1/2" - 3" ± 0,8	26" - 60" ± 3,0 4" - 24" ± 1,5	1/2"-3" + 0,8 4"-24" + 1,5		≤600 ± 0,8 >600 ≤800 ± 1,5 >800 ≤1600 ± 3 >2000 ± 3	
D2	1/2" - 8" ± 0,4 10" - 24" ± 0,8	26"-34" ± 0,8 36"-60" ± 1,3	1/2"-8" + 0,4 10"-24" + 1,5		≤600 ± 0,8 >600 ≤800 ± 1,5 >800 ≤1600 ± 1,5 >1600 ± 3,0	
D3	1/2" - 8" ± 0,8 10" - 24" +1,5 - 0,8	26"-60" ± 1,5	1/2"-8" - 0,8 10"-24" - 1,5		≤600 ± 0,8 >600 ≤1600 ± 1,5 >1600 ± 3,0	
D4	1/2" - 24" - 0,8	26"-60" ± 0,8	1/2"-24" - 0,8		≤600 - 0,8 >600 ≤1000 - 1,5 >1000 ≤2000 - 2,0 >2000 - 3,0	

TSG SPIRAL WOUND GASKETS

For BS 1560 and ASME/ ANSI B 16.5 flanges

Dimensions for SWG (Seaco® TSG SM04 and SM05) for BS 1560 and ASME/ ANSI B 16.5 flanges in accordance with gasket standard BS 3381

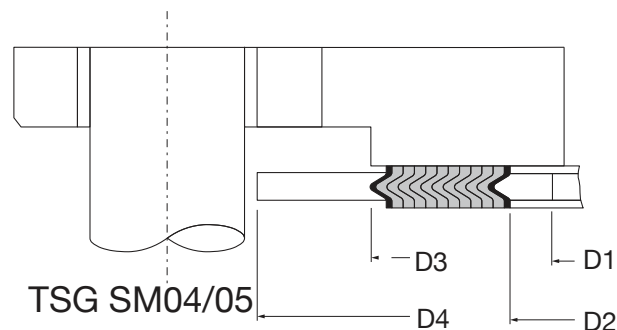


nominal pipe dia	d1 [mm] Pressure class [lbs] 150 - 2500	d2[mm] Pressure class [lbs]			d3[mm] Pressure class [lbs] 150- 2500	d4[mm] Pressure class [lbs]					
		150	300- 1500	2500		150	300	600	900	1500	2500
1/2"	14,3	18,7	18,7	18,7	32,2	47,6	54,0	54,0	63,5	63,5	69,9
3/4"	20,6	26,6	25,0	25,0	40,1	57,2	66,7	66,7	69,9	69,9	76,2
1"	27,0	32,9	31,4	31,4	48,0	66,7	73,0	73,0	79,4	79,4	85,7
1 1/4"	34,9	45,6	44,1	39,3	60,7	76,2	82,6	82,6	88,9	88,9	104,8
1 1/2"	41,3	53,6	50,4	47,2	70,3	85,7	95,3	95,3	98,4	98,4	117,5
2	52,4	69,5	66,3	58,3	86,1	104,8	111,1	111,1	142,9	142,9	146,1
2 1/2"	63,5	82,2	79,0	69,5	98,8	123,8	130,2	130,2	165,1	165,1	168,3
3"	77,8	101,2	94,9	91,7	121,1	136,5	149,2	149,2	168,8	174,6	196,9
4"	103,2	126,6	120,3	117,1	149,6	174,6	181,0	193,7	206,4	209,6	235,0
5"	128,5	153,6	147,2	142,5	178,2	196,9	215,9	241,3	247,7	254,0	279,4
6"	154,0	180,6	174,2	171,1	210,0	222,3	250,8	266,7	288,9	282,6	317,5
8"	203,2	231,4	225,0	215,5	263,9	279,4	308,0	320,7	358,8	352,4	387,4
10"	254,0	286,9	280,6	269,5	317,9	339,7	362,0	400,1	435,0	435,0	476,3
12"	303,2	339,3	333,0	323,5	375,1	409,6	422,3	457,2	498,5	520,7	549,6
14"	342,9	371,1	364,7		406,8	450,9	485,8	492,1	520,7	577,9	
16"	393,7	421,9	415,5		464,0	514,4	539,8	565,2	574,7	641,4	
18"	444,5	475,9	469,5		527,5	549,3	596,9	612,8	638,2	704,9	
20"	495,3	526,7	520,3		578,3	606,4	654,1	682,6	698,5	755,7	
24"	596,9	631,4	625,1		686,2	717,6	774,7	790,6	838,2	901,7	

TSG SPIRAL WOUND GASKETS

For ASME/ANSI B 16.5 flange

Dimensions for SWG (Seaco® TSG SM04 and SM05) for ASME/ANSI B 16.5 flanges in accordance with gasket standard ASME 16.20 (API 601)

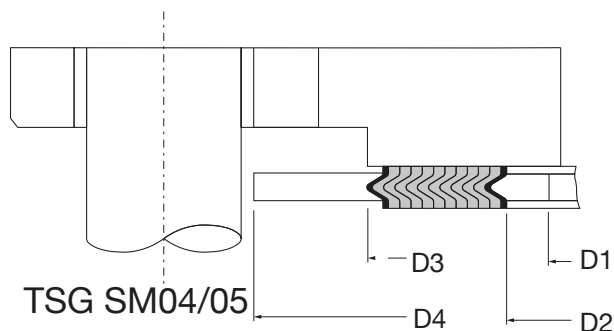


nom. Pipe Dia.	d1 [mm] pressure class [lbs]							d2[mm] pressure class [lbs]							d3[mm]		d4[mm] pressure class [lbs]						
	150	300	400	600	900	1500	2500	150	300	400	600	900	1500	2500	150-600	900-2500	150	300	400	600	900	1500	2500
1/2"	14,2	14,2	14,2	14,2	14,2	14,2	14,2	19,1	19,1	19,1	19,1	19,1	19,1	19,1	31,8	31,8	47,8	54,1	54,1	54,1	63,5	63,5	69,9
3/4"	20,6	20,6	20,6	20,6	20,6	20,6	20,6	25,4	25,4	25,4	25,4	25,4	25,4	25,4	39,6	39,6	57,2	66,8	66,8	66,8	69,9	69,9	76,2
1 "	26,9	26,9	26,9	26,9	26,9	26,9	26,9	31,8	31,8	31,8	31,8	31,8	31,8	31,8	47,8	47,8	66,8	73,2	73,2	73,2	79,5	79,5	85,9
1 1/4"	38,1	38,1	38,1	38,1	33,3	33,3	33,3	47,8	47,8	47,8	47,8	39,6	39,6	39,6	60,5	60,5	76,2	82,6	82,6	82,6	88,9	88,9	104,9
1 1/2"	44,5	44,5	44,5	44,5	41,4	41,4	41,4	54,1	54,1	54,1	54,1	47,8	47,8	47,8	69,9	69,9	85,9	95,3	95,3	95,3	98,6	98,6	117,6
2	55,6	55,6	55,6	55,6	52,3	52,3	52,3	69,9	69,9	69,9	69,9	58,7	58,7	58,7	85,9	85,9	104,9	111,3	111,3	111,3	143,0	143,0	146,0
2 1/2"	66,5	66,5	66,5	66,5	63,5	63,5	63,5	82,6	82,6	82,6	82,6	69,9	69,9	69,9	98,6	98,6	124,0	130,3	130,3	130,3	165,1	165,1	168,4
3"	81,0	81,0	81,0	81,0	81,0	81,0	81,0	101,6	101,6	101,6	101,6	95,3	92,2	92,2	120,7	120,7	136,7	149,4	149,4	149,4	168,4	174,8	196,9
4"	106,4	106,4	106,4	106,4	106,4	106,4	106,4	127,0	127,0	120,7	120,7	120,7	117,6	117,6	149,4	149,4	174,8	181,1	177,8	193,8	206,5	209,6	235,0
5"	131,8	131,8	131,8	131,8	131,8	131,8	131,8	155,7	155,7	147,6	147,6	147,6	143,0	143,0	177,8	177,8	196,9	215,9	212,9	241,3	247,7	254,0	279,4
6"	157,2	157,2	157,2	157,2	157,2	157,2	157,2	182,6	182,6	174,8	174,8	174,8	171,5	171,5	209,6	209,6	222,3	251,0	247,7	266,7	289,1	282,7	317,5
8"	215,9	215,9	209,6	209,6	209,6	206,2	200,2	233,4	233,4	225,6	225,6	222,3	215,9	215,9	263,7	257,3	279,4	308,1	304,8	320,8	358,9	352,6	387,4
10"	268,2	268,2	260,4	260,4	260,4	257,8	247,7	287,3	287,3	274,6	274,6	276,4	266,7	270,0	317,5	311,2	339,9	362,0	358,9	400,1	435,1	435,1	476,3
12"	317,5	317,5	317,5	317,5	314,5	314,5	292,1	339,9	339,9	327,2	327,2	323,9	323,9	317,5	374,7	368,3	409,7	422,4	419,1	457,2	498,6	520,7	549,4
14"	349,3	349,3	349,3	349,3	342,9	339,9		371,6	371,6	362,0	362,0	355,6	362,0		406,4	400,1	450,9	485,9	482,6	492,3	520,7	577,9	
16"	400,1	400,1	400,1	400,1	393,7	387,4		422,4	422,4	412,8	412,8	412,8	406,4		463,6	457,2	514,4	539,8	536,7	565,2	574,8	641,4	
18"	449,3	449,3	449,3	449,3	444,5	438,2		474,7	474,7	469,9	469,9	463,6	463,6		527,1	520,7	549,4	596,9	593,9	612,9	638,3	704,9	
20"	500,1	500,1	500,1	500,1	495,3	489,0		525,5	525,5	520,7	520,7	520,7	520,7		577,9	571,5	606,6	654,1	647,7	682,8	698,5	755,7	
24"	603,3	603,3	603,3	603,3	603,3	577,9		628,7	628,7	628,7	628,7	628,7	628,7		685,8	679,5	717,6	774,7	768,4	790,7	838,2	901,7	

TSG SPIRAL WOUND GASKETS

For ASME/ANSI B 16.5 flanges in accordance with gasket standard ASME 16.20 (API 601)

Spiral wound gaskets (Seaco® TSG SM04 and SM05) for ASME B 16.47 series A (MSS SP-44) flanges in accordance with gasket standard ASME 16.20 (API 601)

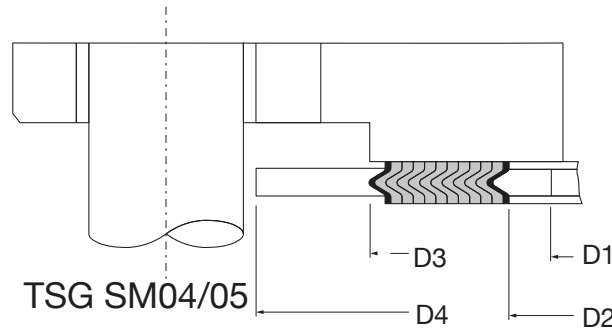


nom. pipe diam.	D1 [mm] pressure class [lbs]					D2[mm] pressure class [lbs]					D3[mm] pressure class [lbs]					D4[mm] pressure class [lbs]				
	150	300	400	600	900	150	300	400	600	900	150	300	400	600	900	150	300	400	600	900
26"	654,1	654,1	660,4	647,7	666,8	673,1	685,8	685,8	685,8	685,8	704,9	736,6	736,6	736,6	736,6	774,7	835,2	831,9	866,9	882,7
28"	704,9	704,9	711,2	698,5	711,2	723,9	736,6	736,6	736,6	736,6	755,7	787,4	787,4	787,4	787,4	831,9	898,7	892,3	914,4	946,2
30"	755,7	755,7	755,7	755,7	774,7	774,7	793,8	793,8	793,8	793,8	806,5	844,6	844,6	844,6	844,6	882,7	952,5	946,2	971,6	1009,7
32"	806,5	806,5	812,8	812,8	812,8	825,5	850,9	850,9	850,9	850,9	860,6	901,7	901,7	901,7	901,7	939,8	1006,6	1003,3	1022,4	1073,2
34"	857,3	857,3	863,6	863,6	863,6	876,3	901,7	901,7	901,7	901,7	911,4	952,5	952,5	952,5	952,5	990,6	1057,4	1054,1	1073,2	1136,7
36"	908,1	908,1	917,7	917,7	920,8	927,1	955,8	955,8	955,8	958,9	968,5	1006,6	1006,6	1006,6	1009,7	1047,8	1117,6	1117,6	1130,3	1200,2
38"	958,9	952,5	952,5	952,5	1009,7	977,9	977,9	971,6	990,6	1035,1	1019,3	1016,0	1022,4	1041,4	1085,9	1111,3	1054,1	1073,2	1104,9	1200,2
40"	1009,7	1003,3	1000,3	1009,7	1060,5	1028,7	1022,4	1025,7	1047,8	1098,6	1070,1	1070,1	1076,5	1098,6	1149,4	1162,1	1114,6	1127,3	1155,7	1251,0
42"	1060,5	1054,1	1051,1	1066,8	1111,3	1079,5	1073,2	1076,5	1104,9	1149,4	1124,0	1120,9	1127,3	1155,7	1200,2	1219,2	1165,4	1178,1	1219,2	1301,8
44"	1111,3	1104,9	1104,9	1111,3	1155,7	1130,3	1130,3	1130,3	1162,1	1206,5	1178,1	1181,1	1181,1	1212,9	1257,3	1276,4	1219,2	1231,9	1270,0	1368,6
46"	1162,1	1152,6	1168,4	1162,1	1219,2	1181,1	1178,1	1193,8	1212,9	1270,0	1228,9	1228,9	1244,6	1263,7	1320,8	1327,2	1273,3	1289,1	1327,2	1435,1
48"	1212,9	1209,8	1206,5	1219,2	1270,0	1231,9	1235,2	1244,6	1270,0	1320,8	1279,7	1286,0	1295,4	1320,8	1371,6	1384,3	1324,1	1346,2	1390,7	1485,9
50"	1263,7	1244,6	1257,3	1270,0		1282,7	1295,4	1295,4	1320,8		1333,5	1346,2	1346,2	1371,6	1435,1	1378,0	1403,4	1447,8		
52"	1314,5	1320,8	1308,1	1320,8		1333,5	1346,2	1346,2	1371,6		1384,3	1397,0	1397,0	1422,4	1492,3	1428,8	1454,2	1498,6		
54"	1358,9	1352,6	1352,6	1378,0		1384,3	1403,4	1403,4	1428,8		1435,1	1454,2	1454,2	1479,6	1549,4	1492,3	1517,7	1555,8		
56"	1409,7	1403,4	1403,4	1428,8		1435,1	1454,2	1454,2	1479,6		1485,9	1505,0	1505,0	1530,4	1606,6	1543,1	1568,5	1612,9		
58"	1460,5	1447,8	1454,2	1473,2		1485,9	1511,3	1505,0	1536,7		1536,7	1562,1	1555,8	1587,5	1663,7	1593,9	1619,3	1663,7		
60"	1511,3	1524,0	1517,7	1530,4		1536,7	1562,1	1568,5	1593,9		1587,5	1612,9	1619,3	1644,7	1714,5	1644,7	1682,8	1733,6		

TSG SPIRAL WOUND GASKETS

For ASME B 16.47 series B (API 605) flanges

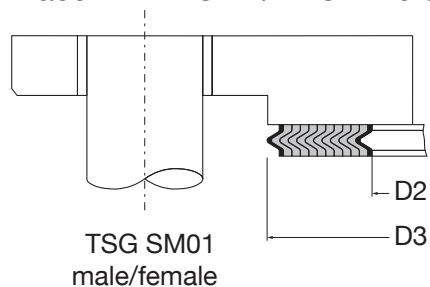
Spiral wound gaskets (Seaco® TSG SM04 and SM05) for ASME B 16.47 series B (API 605) flanges in accordance with gasket standard ASME 16.20 (API 601)



nom. pipe diam.	D1 [mm] pressure class [lbs]					D2 [mm] pressure class [lbs]					D3 [mm] pressure class [lbs]					D4 [mm] pressure class [lbs]				
	150	300	400	600	900	150	300	400	600	900	150	300	400	600	900	150	300	400	600	900
26"	654,1	654,1	654,1	644,7	673,1	673,1	673,1	666,8	663,7	692,2	698,5	711,2	698,5	714,5	749,3	725,4	771,7	746,3	765,3	838,2
28"	704,9	704,9	701,8	692,2	723,9	723,9	723,9	714,5	704,9	743,0	749,3	762,0	749,3	755,7	800,1	776,2	825,5	800,1	819,2	901,7
30"	755,7	755,7	752,6	752,6	787,4	774,7	774,7	765,3	778,0	806,5	800,1	812,8	806,5	828,8	857,3	827,0	886,0	857,3	879,6	958,9
32"	806,5	806,5	800,1	793,8	838,2	825,5	825,5	812,8	831,9	863,6	850,9	863,6	860,6	882,7	914,4	881,1	939,8	911,4	933,5	1016,0
34"	857,3	857,3	850,9	850,9	859,4	876,3	876,3	866,9	889,0	920,8	908,1	914,4	911,4	939,8	971,6	935,0	993,9	962,2	997,0	1073,2
36"	908,1	908,1	898,7	901,7	927,1	927,1	927,1	917,7	939,8	946,2	958,9	965,2	965,2	990,6	997,0	987,6	1047,8	1022,4	1047,8	1124,0
38"	958,9	971,6	952,5	952,5	1009,7	974,6	1009,7	971,6	990,6	1035,1	1009,7	1047,8	1022,4	1041,4	1085,9	1044,7	1098,6	1073,2	1104,9	1200,2
40"	1009,7	1022,4	1000,3	1009,7	1060,5	1022,4	1060,5	1025,7	1047,8	1098,6	1063,8	1098,6	1076,5	1098,6	1149,4	1095,5	1149,4	1127,3	1155,7	1251,0
42"	1060,5	1054,1	1051,1	1066,8	1111,3	1079,5	1079,5	1076,5	1104,9	1149,4	1114,6	1117,6	1127,3	1155,7	1200,2	1146,3	1200,2	1178,1	1219,2	1301,8
44"	1111,3	1124,0	1104,9	1111,3	1155,7	1124,0	1162,1	1130,3	1162,1	1206,5	1165,4	1200,0	1181,1	1212,9	1257,3	1197,1	1251,0	1231,9	1270,0	1368,6
46"	1162,1	1178,1	1168,4	1162,1	1219,2	1181,1	1216,2	1193,8	1212,9	1270,0	1224,0	1254,3	1244,6	1263,7	1320,8	1255,8	1317,8	1289,1	1327,2	1435,1
48"	1212,9	1200,2	1206,5	1219,2	1270,0	1231,9	1231,9	1244,6	1270,0	1320,8	1270,0	1270,0	1295,4	1320,8	1371,6	1306,6	1368,6	1346,2	1390,7	1485,9
50"	1263,7	1267,0	1257,3	1270,0		1282,7	1317,8	1295,4	1320,8		1325,6	1355,9	1346,2	1371,6	1357,4	1419,4	1403,4	1447,8		
52"	1314,5	1317,8	1308,1	1320,8		1333,5	1368,6	1346,2	1371,6		1376,4	1406,7	1397,0	1422,4	1408,2	1470,2	1454,2	1498,6		
54"	1365,3	1346,2	1352,6	1378,0		1384,3	1384,3	1403,4	1428,8		1422,4	1422,4	1454,2	1479,6	1463,8	1530,4	1517,7	1555,8		
56"	1412,7	1428,8	1403,4	1428,8		1435,1	1479,6	1454,2	1479,6		1472,2	1524,0	1505,0	1530,4	1514,6	1593,9	1568,5	1612,9		
58"	1463,5	1484,4	1454,2	1473,2		1485,9	1535,2	1505,0	1536,7		1522,5	1573,3	1555,8	1587,5	1579,6	1655,8	1619,3	1663,7		
60"	1514,3	1505,0	1517,7	1530,4		1536,7	1536,7	1568,5	1593,9		1573,3	1574,8	1619,3	1644,7	1630,4	1706,6	1682,8	1733,6		

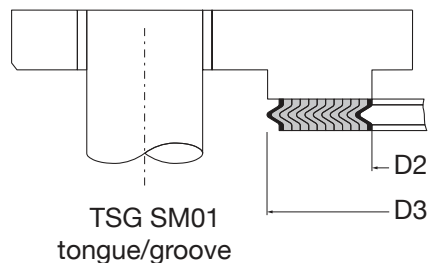
TSG SPIRAL WOUND GASKETS

Seaco TSG® SM01, 150lbs-1500lbs
for flange shape male/female
in acc. with ASME/ANSI B16.5



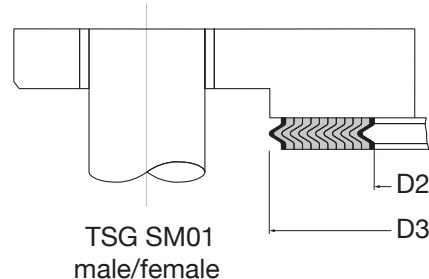
Nom pipe dia.	Narrow D3 [mm]	Wide	
		D3 [mm]	D2 [mm]
1/2"	18	21	35
3/4"	24	27	43
1"	30	33	51
1 1/4"	38	42	64
1 1/2"	44	48	73
2"	57	60	92
2 1/2"	68	73	105
3"	84	89	127
3 1/2"	97	102	140
4"	110	114	157
5"	137	141	186
6"	162	168	216
8"	213	219	270
10"	267	273	324
12"	318	324	381
14"	349	356	413
16"	400	406	470
18"	451	457	535
20"	502	510	585
24"	603	610	690

Seaco TSG® SM01, 150lbs-1500lbs
for flange shape tongue/groove
in acc. with ASME/ANSI B16.5



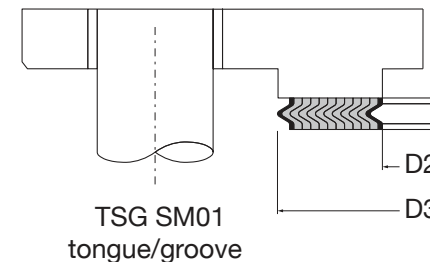
Nom. pipe dia.	D2 [mm]	Narrow D3 [mm]	Wide D3 [mm]
1/2"	25	35	35
3/4"	33	43	43
1"	38	48	51
1 1/4"	48	57	64
1 1/2"	54	64	73
2"	73	83	92
2 1/2"	86	95	105
3"	108	118	127
3 1/2"	121	130	140
4"	132	145	157
5"	160	173	186
6"	190	203	216
8"	238	254	270
10"	286	305	324
12"	343	36	381
14"	375	394	413
16"	425	448	470
18"	489	511	535
20"	535	559	585
22"	591	616	641
24"	640	667	690

Seaco TSG® SM01, PN 10-100 bar
for flange shape male/female
DIN 2692 in acc. with DIN 2692



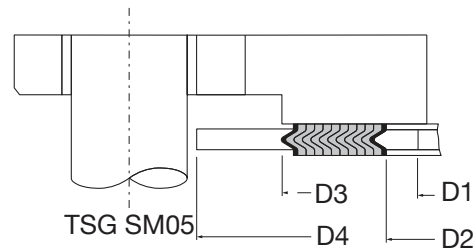
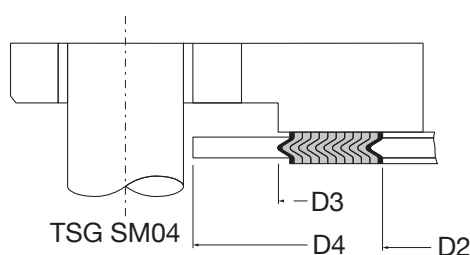
Nom. pipe diam.	d2 [mm]	d3 [mm]
10	18	34
15	22	39
20	28	50
25	35	57
32	43	65
40	49	75
50	61	87
65	77	109
80	90	120
100	115	149
125	141	175
150	169	203
175	195	233
200	220	259
250	274	312
300	325	363
350	368	421
400	420	473
500	520	575
600	620	675
700	720	777
800	820	882
900	920	987
1000	1020	1091

Seaco TSG® SM01, PN 10-100 bar
for flange shape tongue/groove
DIN 2512 in acc. with DIN 2691



Nom. pipe diam.	D2 [mm]	D3 [mm]
4-6	20	30
8	22	32
10	24	34
15	29	39
20	36	50
25	43	57
32	51	65
40	61	75
50	73	87
65	95	109
80	106	120
100	129	149
125	155	175
150	183	203
175	213	233
200	239	259
250	292	312
300	343	363
350	395	421
400	447	473
500	549	575
600	649	675
700	751	777
800	856	882
900	961	987
1000	1062	1092

Seaco TSG® SM04 and SM05) for DIN flanges



DN		PN10				PN16		PN25		PN40		PN63			PN100			PN160		
mm	inch	D0	D1	D2	D3	D0/D1/D2	D3	D0/D1/D2	D3	D0/D1/D2	D3	D0/D1	D2	D3	D0/D1	D2	D3	D0/D1	D2	D3
10	1 / 4	18	24	34	46	idem PN10	46	idem PN10	46	idem PN10	46	idem PN10	34	56	idem PN10	idem PN64	61	idem PN10	idem PN64	61
20	3/4	28	34	46	61		61		61		61		-	-			-			-
25	1	35	41	53	71		71		71		71		53	82			82			82
32	1 1/4	43	49	61	82		82		82		82		-	-			-			-
40	1 1/2	50	56	68	92		92		92		92		68	103			103			103
50	2	61	70	86	107		107		107		107		86	113			119			119
65	2 1/2	77	86	102	127		127		127		127		106	137			143			143
80	3	90	99	115	142		142		142		142		119	148			154			154
100	4	115	127	143	162		162		168		168		147	174			180			180
125	5	140	152	172	192		192		194		194		176	210			217			217
150	6	167	179	199	218		218		224		224		203	247			257			257
200	8	216	228	248	273		273		284		290		252	309			324			324
250	10	267	279	303	327		329		340		352		307	364			391			388
300	12	318	330	354	377		384		400		417		358	424			458			458
350	14	360	376	400	437		444		457		474		404	486			512			-
400	16	410	422	450	488		495		514		546		456	543			572			-
500	20	510	522	550	593		617		624		628		556	657			704			-
600	24	610	622	650	695		734		731		747		656	764			813			-
700	28	710	722	756	810		804		833		852		762	879			950			-
800	32	810	830	864	917		911		942		974		870	988			-			-
900	36	910	930	964	1017		1011		1042		1084		970	1108			-			-
1000	40	1010	1030	1074	1124		1128		1154		1194		1080	-			-			-

Installation procedure

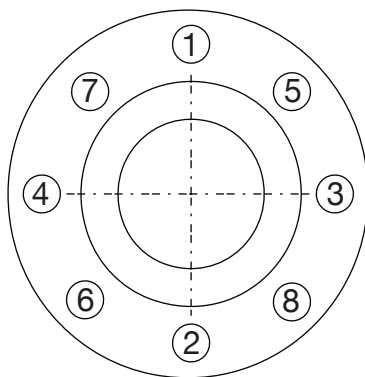
Please pay attention to the installation procedure.

Most of the spiral wound gasket's failures is breach of the recommended assembly procedure.

- Clean contact surfaces of the flanges and check them if they are not damaged.
- Inspect and clean bolts, nuts and washers.
- Make certain the parallelism of the contact surfaces.
- Put spiral wound gasket right in the centre
- Tight the bolts approximately 50% of the final torque according to the sequence in the figure (repeat then 75%, 100%).

If the correct tighten sequence is not followed the gasket will be damaged and not reach required sealing performance.

- We do not recommend bolts un-tightening and re-tightening.
- Pay extra attention during installation spiral wound gaskets type Seaco® TSG SM01



Loading calculation of gasket compresibility

If a gasket has to reach its performance it must be loaded properly.

Filler	Q [N/mm ²] at 20° C			M	GASKETS WITH OUTER RINGS				
					Qt [N/mm ²]				
					100°C	200°C	300°C	400°C	500°C

Graphite	50	95	180	2	140	130	120	110	100
PTFE	50	80	130	1,8	105	90	-	-	-

Filler	Q [N/mm ²] at 20° C			M	GASKETS WITH OUTER RINGS				
					Qt [N/mm ²]				
					100°C	200°C	300°C	400°C	500°C

Graphite	50	122	300	2	180	170	160	150	140
PTFE	0	110	250	1,8	150	130	-	-	-

Example

Gasket type SeacoTSG® OR/IR, Temperature 400°C, DN = 100 mm, PN = 40 Bar, Filler = graphite

Assembling stress

recommended $Q_{rec} = 122 \text{ N/mm}^2$, Minimal¹⁾, $Q_{min}=50 \text{ N/mm}^2$, Maxima²⁾
 $Q_{max}=300 \text{ N/mm}^2$

¹⁾ Gasket sealing is started, ²⁾ Gasket damage will occur

Gasket loading calculation

Gasket service area : $A = (d_{32}-d_{22}).\pi/4$ $A = (1402-1222).\pi/4 = 4716 \text{ mm}^2$

Bolt loading

Recommended.: $F_{rec}=Q_{rec}.A$, $F_{rec} = 122.4716=575352 \text{ N}$,
 Minimal: $F_{min}=Q_{min}.A$, $F_{min}= 50.4716=235800 \text{ N}$,
 Maximal: $F_{max}=Q_{max}.A$, $F_{max}= 300.4716=1\,414\,800 \text{ N}$

More precise calculation

Sealing coefficient $m, M=2$
Bolt loading minimal $F_m[N]$ $F_m=m.P.A$ $F_m=2.4.4716=37728\text{ N}$
Medium pressure $P [N/mm^2]$ $P=4\text{ N/mm}^2$
Separation force $F_p[N]$ $F_p=d22.\pi.P/4$ $F_p=1202.\pi.4/4=45\,239\text{ N}$
Minimal total loading $F_{cmin}[N]$ $F_{cmin}=F_m+F_p$ $F_{cmin}=37728+45\,239=82967\text{ N}$

Assembling stress

According to temperature: $Q_t[N/mm^2]$, $Q_t=150\text{ N/mm}^2$
Bolt loading maximal $F_{tmax}[N]$ $F_{tmax}=Q_t.A$ $F_{tmax}=150*4716=707400\text{ N}$
Bolt loading final $F [N]$ $F < F_{max}, F_{tmax}$

$F > F_{min}, F_{cmin}$ $F < 414800\text{ N}, 707400\text{ N}$
 $F < 707400\text{ N}$
 $F > 235800\text{ N}, 82967\text{ N}$
 $F > 235800\text{ N}$
 $235800\text{ N} < F < 707400\text{ N}$
 $F = 471000\text{ N}$

Number of bolts x See table of bolts for DIN flanges 8 pc M20
Loading per 1 bolt $F_{\check{s}}[N]$ $F_{\check{s}}=F/x$ $471000/8=58875\text{ N}$
Bolt torque $M [N.m]$ See table of bolt torque and bolt loading
Chosen quality 5,6 (5D) * $M=184\text{ N.m}$

* It could be used also bolt quality 6.9(6G). In this case the bolt torque equals $M = 335\text{ N.m}$
(Tight for 60% M!)

Bolt numbers and sizes for ASME flanges (12" - 32")

Nomi. Diam.	Pressure class						
	150 lbs	300 lbs	400 lbs	600 lbs	900 lbs	1500 lbs	2500 lbs
1/2"	4x1/2"	4x1/2"	4x5/8"	4x1/2"	4x3/4"	4x3/4"	4x3/4"
3/4"	4x1/2"	4x5/8"	4x5/8"	4x5/8"	4x3/4"	4x3/4"	4x3/4"
1"	4x1/2"	4x5/8"	4x5/8"	4x5/8"	4x7/8"	4x7/8"	4x7/8"
1 1/4"	4x1/2"	4x5/8"	4x5/8"	4x5/8"	4x7/8"	4x7/8"	4x1"
1 1/2"	4x1/2"	4x3/4"	4x3/8"	4x3/4"	4x1"	4x1"	4x1 1/8"
2"	4x5/8"	5x5/8"	8x5/8"	5x5/8"	8x7/8"	8x7/8"	8x1"
2 1/2"	4x5/8"	8x3/4"	8x3/4"	8x3/4"	8x1"	8x1"	8x1 1/8"
3"	4x5/8"	8x3/4"	8x3/4"	8x3/4"	8x7/8"	8x1 1/8"	8x1 1/4"
3 1/2"	8x5/8"	8x3/4"	8x7/8"	8x7/8"			
4"	8x5/8"	8x3/4"	8x7/8"	8x7/8"	8x1 1/8"	8x1 1/4"	8x1 1/2"
5"	8x3/4"	8x3/4"	8x7/8"	8x1"	8x1 1/4"	8x1 1/2"	8x1 3/4"
6"	8x3/4"	12x3/4"	12x7/8"	12x1"	12x1 1/8"	12x1 3/8"	8x2"
8"	8x3/4"	12x7/8"	12x1"	12x1 1/8"	12x1 3/8"	12x1 5/8"	12x2"
10"	12x7/8"	16x1"	16x1 1/8"	16x1 1/4"	16x1 3/8"	12x1 7/8"	12x2 1/2"
12"	12x7/8"	16x1 1/8"	16x1 1/4"	20x1 1/4"	20x1 3/8"	16x2"	12x2 3/4"
14"	12x1"	20x1 1/8"	20x1 1/4"	20x1 3/8"	20x1 1/2"	16x2 1/4"	
16"	16x1"	20x1 1/4"	20x1 3/8"	20x1 1/2"	20x1 5/8"	16x2 1/2"	
18"	16x1 1/8"	24x1 1/4"	24x1 3/8"	20x1 5/8"	20x1 7/8"	16x2 3/4"	
20"	20x1 1/8"	24x1 1/4"	24x1 1/2"	24x1 5/8"	20x2"	16x3"	
22"	20x1 1/4"	24x1 1/2"	24x1 5/8"	24x1 3/4"			
24"	20x1 1/4"	24x1 1/2"	24x1 3/4"	24x1 7/8"	20x2 1/2"	16x3 1/2"	
26"	24x1 1/4"	26x1 5/8"	28x1 3/4"	28x1 7/8"	20x2 3/4"		
28"	28x1 1/4"	28x1 5/8"	28x1 7/8"	28x2"	20x3"		
30"	28x1 1/4"	28x1 3/4"	28x2"	28x2"	20x3"		
32"	28x1 1/2"	28x1 7/8"	28x2"	28x2 1/4"	20x3 1/4"		

Bolt numbers and sizes for ASME flanges (34" - 60")

Nomi. Diam.	Pressure class				
	150 lbs	300 lbs	400 lbs	600 lbs	900 lbs
34"	32x1 1/2"	28x1 7/8"	28x2"	28x2 1/4"	20x3 1/2"
36"	32x1 1/2"	32x2"	32x2"	28x2 1/2"	20x3 1/2"
38"	32x1 1/2"	32x1 1/2"	32x1 3/4"	28x2 1/4"	20x3 1/2"
40"	36x1 1/2"	32x1 5/8"	32x1 7/8"	32x2 1/4"	24x3 1/2"
42"	36x1 1/2"	36x2"	32x2 1/2"	28x2 3/4"	24x3 1/2"
44"	40x1 1/2"	32x1 3/4"	32x2"	32x2 1/2"	24x3 3/4"
46"	40x1 1/2"	28x1 7/8"	36x2"	32x2 1/2"	24x4"
48"	44x1 1/2"	32x1 7/8"	28x2 1/4"	32x2 3/4"	24x4"
50"	44x1 3/4"	32x2"	32x2 1/4"	28x3"	
52"	44x1 3/4"	32x2"	32x2 1/4"	32x3"	
54"	44x1 3/4"	28x2 1/4"	28x2 1/2"	32x3"	
56"	48x1 3/4"	32x2 1/4"	32x2 1/2"	32x3 1/4"	
58"	48x1 3/4"	28x2 1/4"	32x2 1/2"	32x3 1/4"	
60"	52x1 3/4"	32x2 1/4"	32x2 3/4"	28x3 1/2"	

Bolt numbers and sizes for DIN flanges

DN/PN	1 - 2,5	6	10	16	25	40	64	100	160	250	320	400
6		4 M 10				4 M 10						
8		4 M 10				4 M 10						
10		4 M 10				4 M 12	4 M 12	4 M 12	4 M 12	4 M 16	4 M 16	4 M 16
15		4 M 10				4 M 12	4 M 12	4 M 12	4 M 12	4 M 16	4 M 16	4 M 20
20		4 M 10				4 M 12						
25		4 M 10				4 M 12	4 M 16	4 M 16	4 M 16	4 M 20	4 M 20	4 M 24
32		4 M 12				4 M 16						
40		4 M 12				4 M 16	4 M 20	4 M 20	4 M 20	4 M 24	4 M 24	4 M 27
50		4 M 12				4 M 16	4 M 20	4 M 24	4 M 24	8 M 24	8 M 24	8 M 27
65		4 M 12		4 M 16		8 M 16	8 M 20	8 M 24	8 M 24	8 M 24	8 M 27	8 M 30
80		4 M 16		8 M 16		8 M 16	8 M 20	8 M 24	8 M 24	8 M 27	8 M 27	8 M 30
100		4 M 16		8 M 16		8 M 20	8 M 24	8 M 27	8 M 27	8 M 30	8 M 33	8 M 36
125		8 M 16		8 M 16		8 M 24	8 M 27	8 M 30	8 M 30	12 M 30	12 M 33	12 M 36
150		8 M 16		8 M 20		8 M 24	8 M 30	12 M 30	12 M 30	12 M 33	12 M 36	12 M 39
175			8 M 20	12 M 24		12 M 27	12 M 30	12 M 30	12 M 33	12 M 36	12 M 39	12 M 45
200		8 M 16	8 M 20	12 M 20	12 M 24	12 M 27	12 M 33	12 M 33	12 M 33	12 M 39	16 M 39	16 M 45
250		12 M 16	12 M 20	12 M 24	12 M 27	12 M 30	12 M 33	12 M 36	12 M 39	16 M 45	16 M 48	
300		12 M 16	12 M 20	12 M 24	16 M 27	16 M 30	16 M 39	16 M 39	16 M 39	16 M 48		
350		12 M 20	16 M 20	16 M 24	16 M 30	16 M 33	16 M 36	16 M 45				
400		16 M 20	16 M 24	16 M 27	16 M 33	16 M 36	16 M 39	16 M 45				
450		16 M 20	20 M 24	20 M 27		20 M 36						
500		20 M 20	20 M 24	20 M 30	20 M 33	20 M 39	20 M 45	20 M 52				
600		20 M 24	20 M 27	20 M 33	20 M 36	20 M 45	20 M 52	20 M 56				
700		24 M 24	24 M 27	24 M 33	24 M 39	24 M 45	24 M 52	24 M 64				
800		24 M 27	24 M 30	24 M 36	24 M 45	24 M 52	24 M 56					
900		24 M 27	28 M 30	28 M 36	28 M 45	28 M 52	28 M 56					
1000		28 M 27	28 M 33	28 M 39	28 M 52	28 M 52	29 M 64					
1200	32 M 27	32 M 30	32 M 36	32 M 45	32 M 52	32 M 56	32 M 72					
1400	36 M 27	36 M 33	36 M 39	36 M 45	36 M 56	36 M 56						
1600	40 M 27	40 M 33	40 M 45	40 M 52	40 M 56	40 M 64						
1800	44 M 27	44 M 36	44 M 45	44 M 52	44 M 64							
2000	48 M 27	48 M 39	48 M 45	48 M 56	48 M 64							
2200	52 M 30	52 M 39	52 M 52	52 M 56								
2400	56 M 30	56 M 39	56 M 52									
2600	60 M 30	60 M 45	60 M 52									
2800	64 M 33	64 M 45	64 M 52									
3000	68 M 33	68 M 45	68 M 56									

Bolt loading F and bolt torque M

Bolt Marking	3.6 (4D)		5.6 (5D)		6.9 (6G)		8.8 (8G)		10.9 (10K)		12.9 (12K)	
	F [N]	M [N.m]	F [N]	M [N.m]	F [N]	M [N.m]	F [N]	M [N.m]	F [N]	M [N.m]	F [N]	M [N.m]
M10	8640	17	115001	22	22200	43	26300	51	37000	72	44400	87
M12	12600	29	16800	39	32400	75	38400	89	54000	125	64800	150
M14	17300	46	23100	62	44500	119	52700	141	741000	198	89000	240
M16	23800	71	31700	95	61200	182	72600	215	102000	305	122500	365
M18	28900	97	38600	130	74400	250	88200	295	124000	420	149000	500
M20	37200	138	49600	184	95700	355	113500	420	159500	590	191500	710
M22	46500	186	62000	250	119500	480	142000	570	190500	800	239000	960
M24	53600	235	71400	315	138000	610	163500	725	230000	1020	276000	1220
M27	70600	350	94100	470	181500	905	215000	1070	302000	1510	363000	1810
M30	85700	475	114500	635	220000	1230	261000	1450	367000	2050	441000	2450
M33	10700	645	142500	865	275000	1660	326000	1970	458000	2770	550000	3330
M36	125500	830	167500	1110	323000	2140	382000	2530	538000	3560	645000	4280
M39	151000	1080	201000	1440	388000	2770	460000	3290	646000	4620	776000	5550
M42	172500	1330	230000	1780	444000	3430	526000	4070	740000	5720	888000	6860
M45	202000	1670	270000	2220	520000	4290	616000	5090	867000	7150	1040000	8580
M48	227000	2020	303000	2690	585000	5180	693000	6140	975000	8640	1170000	10350
M52	273000	2590	364000	3460	702000	6670	832000	7900	1170000	11100	1405000	13350

Values correspond for friction coefficient 0,14 e.g. usual bolts with slightly oiled friction surfaces.

ASME B16.20 color code chart Spiral wound gasket metallic windings and non-metallic fillers

Metal	OD ring color		Metal	OD ring color	
304 stainless steel	Yellow		HastelloyC [®]	Beige	
316L stainless steel	Green		Incoloy800 [®]	White	
317L stainless steel	Maroon				
321 stainless steel	Turquoise				
347 stainless steel	Blue				
Monel [®]	Orange		Material	Stripe color	
Nickel	Red		Flexible graphite	Gray	
Titanium	Purple		PTFE	White	
Alloy 20	Black		Ceramic	Lt. Green	
Inconel600 [®]	Gold		Verdicarb (mica graphite)	Pink	
HastelloyB [®]	Brown				

Waranty exclusion

In view of the variety of different installation and operation conditions as well as application and process engineering options, the information given in this datasheet can only provide approximate guidance and cannot be used as basis for warranty claims.